Assignment 4 — RDBMS vs NoSQL

Project: College Management System | Database: college_management_system | SQL dialect: MySQL 8.0

Based on the seven-table college design supplied for this project. Example records and expected outputs are illustrative, not screenshots from MySQL Workbench.

Relational database versus NoSQL

Dimension	Relational (MySQL)	NoSQL (e.g., MongoDB)
Data model	Fixed tables, rows, joins	Documents / key-value / graph / wide-column
Scalability	Often vertical; also replication/sharding	Often designed for horizontal distribution
Consistency	ACID transactions; strong integrity rules	Varies by product/configuration; may offer transactions
Performance	Strong joins and structured reports	Good for flexible documents and access patterns
Schema	Explicit DDL and constraints	Flexible or application-defined structure
Typical use	Student records, enrollments, grades	Activity events, content, large flexible logs

Why organizations choose differently

A college needs accurate enrollment and result records, referential integrity, uniqueness, and multi-table reports. MySQL is a suitable primary system of record. A document store can complement it for student activity feeds, heterogeneous event logs or optional profile metadata.

Real-world examples

Banking and college registration systems often favor relational transactions to preserve balances or enrollment relationships. Content catalogs and event-logging applications may choose document databases when fields vary across records. Many organizations use both technologies for different workloads.

Project decision

Choose MySQL 8.0 for this College Management System because students, departments, courses, enrollments and results have clear structured relationships. NoSQL is an optional secondary system, not a replacement for integrity-critical academic records.